# Sonning moduli. Modul qatnashgan tenglama va tengsizliklar

Termometr −6 °C ni ham, 6 °C ni ham ko'rsatishi mumkin. Ikkala son ham noldan 6 birlik uzoqda turibdi. Sonning noldan qanchalik uzoqligini uning moduli ko'rsatadi.

1-QISM

## Mavzu

TA'RIF · SONNING MODULI

1) **Musbat sonning moduli shu sonning o'ziga teng.**

2) **Manfiy sonning moduli unga qarama-qarshi songa teng.**

3) **Nolning moduli nolga teng:** $|0| = 0$.

Shunday qilib, son modulining ta'rifi formula yordamida qisqacha shunday yoziladi:

$$|a| = \begin{cases} a, & \text{agar } a \geq 0 \text{ bo'lsa;} \\ -a, & \text{agar } a < 0 \text{ bo'lsa.} \end{cases}$$

1-MISOL · SONNING MODULI

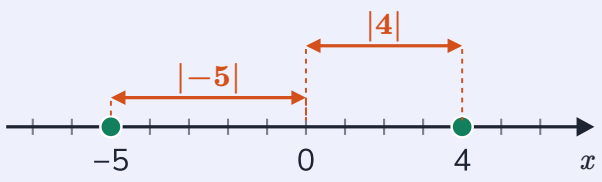
1. $|14| = 14$ — musbat son, moduli o'ziga teng.
2. $|-9| = 9$ — manfiy son, uning qarama-qarshisi 9. Xuddi shunday $\left|-\frac{4}{7}\right| = \frac{4}{7}$, $|-0{,}8| = 0{,}8$.

2-MISOL · IFODANING QIYMATI

1. $|-6| + |2{,}5| - |-1{,}5| = 6 + 2{,}5 - 1{,}5 = 7$.
2. $x = 3$ bo'lsa: $|x - 8| = |3 - 8| = |-5| = 5$.

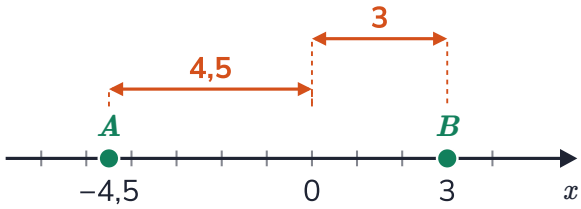
TA'RIF · MODULNING GEOMETRIK MA'NOSI

**$|a|$ — $a$ sonning moduli, geometrik nuqtayi nazardan, 0 nuqtadan $a$ sonni tasvirlovchi nuqtagacha bo'lgan masofadir.**



**1-MISOL · MASOFANI TOPISH**

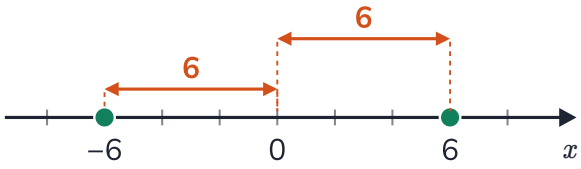
$A$ nuqta $-4{,}5$ sonni, $B$ nuqta 3 sonni tasvirlaydi. Qaysi biri 0 ga yaqinroq?




1. $|-4{,}5| = 4{,}5$, $|3| = 3$. $3 < 4{,}5$, demak $B$ nuqta 0 ga yaqinroq.

**2-MISOL · NUQTANI TOPISH**

Son o'qida 0 dan 6 birlik uzoqlikda qaysi nuqtalar yotadi?




1. 0 dan o'ngga 6 birlik — 6, chapga 6 birlik — $-6$. Javob: $6$ va $-6$.

**USUL · MODUL BELGISI OSTIDA NOMA'LUM QATNASHGAN TENGLAMA**

1. Modul ichidagi ifoda $\geq 0$ bo'lsin: modul belgisini tushirib qoldiramiz va hosil bo'lgan tenglamani yechamiz.
2. Modul ichidagi ifoda $< 0$ bo'lsin: ifodani o'ng tomondagi sonning qarama-qarshisiga tenglab yechamiz.
3. Ikkala holda topilgan sonlar — tenglamaning ildizlari.

**1-MISOL · MODUL ICHIDA FAQAT NOMA'LUM**

$|x| = 9$ tenglamani yeching.

1. $x \geq 0$ bo'lsin: $|x| = x$, tenglama $x = 9$.
2. $x < 0$ bo'lsin: $|x| = -x$, tenglama $-x = 9$, bundan $x = -9$.
3. Javob: $x_1 = 9$, $x_2 = -9$. Ikkalasi ham 0 dan 9 birlik uzoqda.

**2-MISOL · MODUL ICHIDA IFODA**

$|2x - 5| = 7$ tenglamani yeching.

1. $2x - 5 \geq 0$ bo'lsin: $2x - 5 = 7$, $2x = 12$, $x = 6$.
2. $2x - 5 < 0$ bo'lsin: $2x - 5 = -7$, $2x = -2$, $x = -1$.
3. Tekshiramiz: $|12 - 5| = 7$, $|-2 - 5| = 7$. Javob: $x_1 = 6$, $x_2 = -1$.

**QOIDA · «≤» BELGILI MODULLI TENGSIZLIK**

Ushbu $|x| \le a$, bunda $a > 0$, tengsizlikni 0 nuqtadan $a$ dan katta bo'lmagan masofada yotuvchi barcha $x$ nuqtalar, ya'ni $[-a;\ a]$ kesmaning nuqtalari qanoatlantiradi.

$[-a;\ a]$ kesma — ushbu $-a \le x \le a$ tengsizlikni qanoatlantiruvchi $x$ sonlar to'plami.

**Demak, $|x| \le a$ tengsizlik $-a \le x \le a$ qo'sh tengsizlikning ayni o'zini bildiradi, bunda $a > 0$.**




**1-MISOL · MODUL ICHIDA FAQAT NOMA'LUM**

$|x| < 4$ tengsizlikni yeching.

1. 0 dan 4 birlikdan kam masofadagi nuqtalar: $-4 < x < 4$.
2. $-4$ va 4 kirmaydi ($|4| = 4$, 4 esa 4 dan kichik emas), shuning uchun ular bo'sh doiracha.




**2-MISOL · MODUL ICHIDA IFODA**

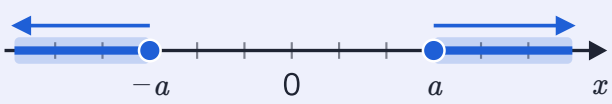
$|2x - 1| \le 5$ tengsizlikni yeching.

1. Qo'sh tengsizlik: $-5 \le 2x - 1 \le 5$. U ushbu sistemaning xuddi o'zi: $\begin{cases} 2x - 1 \le 5, \\ 2x - 1 \ge -5. \end{cases}$
2. Birinchisidan $x \le 3$, ikkinchisidan $x \ge -2$. Javob: $-2 \le x \le 3$.




**QOIDA · «≥» BELGILI MODULLI TENGSIZLIK**

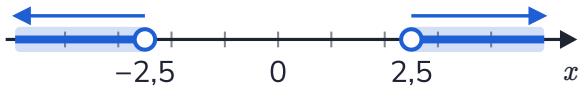
Ushbu $|x| \ge a$, bunda $a > 0$, tengsizlikni 0 nuqtadan $a$ dan kichik bo'lmagan masofada yotuvchi barcha $x$ nuqtalar to'plami, ya'ni $x \ge a$ va $x \le -a$ nurlarning nuqtalari qanoatlantiradi.



**1-MISOL · MODUL ICHIDA FAQAT NOMA'LUM**

$|x| > 2{,}5$ tengsizlikni yeching.

1. 0 dan 2,5 birlikdan uzoqroq nuqtalar ikki tomonda yotadi: $x < -2{,}5$ yoki $x > 2{,}5$.




**2-MISOL · MODUL ICHIDA IFODA**

$|x + 3| \geq 4$ tengsizlikni yeching.

1. $x + 3 \geq 0$ bo'lsin: $x + 3 \geq 4$, $x \geq 1$.
2. $x + 3 < 0$ bo'lsin: $-(x + 3) \geq 4$, ya'ni $x + 3 \leq -4$, $x \leq -7$.
3. Javob: $x \leq -7$ yoki $x \geq 1$.




«$\leq$» bo'lsa, yechim 0 ga yaqin nuqtalar — bitta kesma. «$\geq$» bo'lsa, 0 dan uzoq nuqtalar — ikkita nur.



**QOIDA · O'NG TOMONDA NOL YOKI MANFIY SON**

**Agar $|x| \leq a$ tengsizlikda $a$ son nolga teng bo'lsa, u holda tengsizlik $x = 0$ dan iborat birgina (yagona) yechimga ega bo'ladi; bordi-yu, agar $a < 0$ bo'lsa, u holda tengsizlik yechimlarga ega bo'lmaydi.**

**Agar $|x| \geq a$ tengsizlikda $a$ son noldan kichik yoki unga teng bo'lsa, u holda istalgan son uning yechimi bo'ladi.**

**1-MISOL · «≤» BELGI**

1. $|x - 2| \leq 0$ faqat $x - 2 = 0$ da bajariladi: $x = 2$ — yagona yechim.
2. $|3x + 1| \leq -4$ — **yechim yo'q**: modul manfiy bo'lmaydi.

**2-MISOL · «≥» BELGI**

1. $|5 - x| \geq -2$ — **istalgan son**: modul doim $\geq 0$, demak $-2$ dan katta.
2. $|x + 7| \geq 0$ — ham **istalgan son**.

2-QISM

# Misollar ishlash

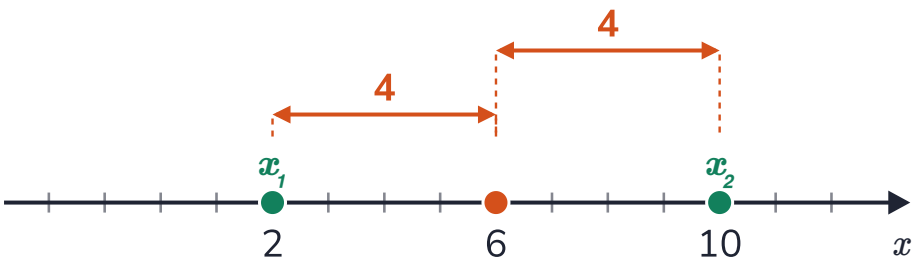


Tengsizlikni manfiy songa bo'lsangiz, belgini teskarisiga almashtiring: «$<$» o'rniga «$>$», «$\leq$» o'rniga «$\geq$».

## 1-MISOL · NOMA'LUM OLDIDA MINUS

$|6 - x| = 4$ tenglamani yeching.

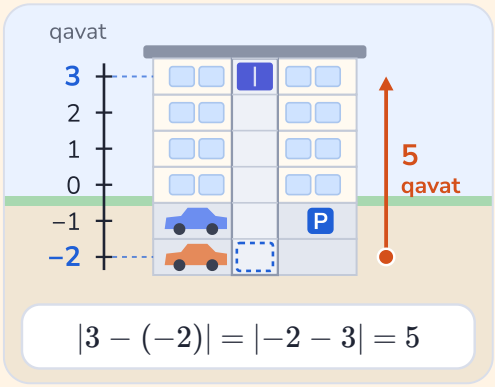
1. $6 - x \geq 0$ bo'lsin: $6 - x = 4$, bundan $x = 2$.
2. $6 - x < 0$ bo'lsin: $6 - x = -4$, bundan $x = 10$.
3. Son o'qida tekshiramiz: $|6 - x|$ — $x$ va 6 nuqtalar orasidagi masofa. 6 dan 4 birlik uzoqlikda 2 va 10 yotadi.
4. Javob: $x_1 = 2$, $x_2 = 10$.




**DIQQAT**

$|x - a|$ — son o'qida $x$ va $a$ nuqtalar orasidagi masofa. Masalan, $|7 - 3| = 4$ va $|3 - 7| = 4$: 3 va 7 nuqtalar orasida 4 birlik bor.

Hayotda: lift $-2$-qavatdan 3-qavatga chiqsa ham, qaytib tushsa ham 5 qavat yuradi (rasmda).



**2-MISOL · KASR KOEFFITSIENTLI TENGLAMA**

$\left|\frac{1}{2}x+\frac{1}{3}\right|=\frac{5}{6}$ tenglamani yeching.

1. Modul ichidagi ifoda $\geq 0$: $\frac{1}{2}x+\frac{1}{3}=\frac{5}{6}$. Ayiramiz: $\frac{5}{6}-\frac{1}{3}=\frac{1}{2}$, ya'ni $\frac{1}{2}x=\frac{1}{2}$, $x=1$.
2. Modul ichidagi ifoda $<0$: $\frac{1}{2}x+\frac{1}{3}=-\frac{5}{6}$. Bundan $\frac{1}{2}x=-\frac{5}{6}-\frac{1}{3}$, ya'ni $\frac{1}{2}x=-\frac{7}{6}$.
3. Ikkala tomonni 2 ga ko'paytiramiz: $x=-\frac{7}{3}=-2\frac{1}{3}$.
4. Javob: $x_1=1$, $x_2=-2\frac{1}{3}$.

**3-MISOL · BUTUN YECHIMLAR**

$|7-2x|<3$ tengsizlikni yeching va uning butun yechimlarini toping.

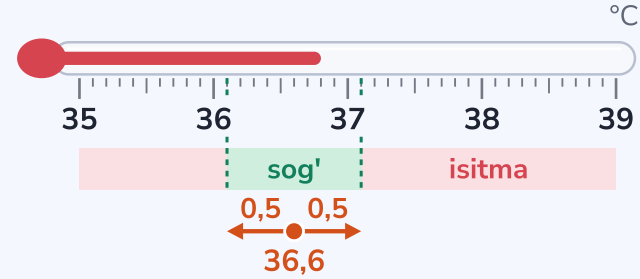
1. Qo'sh tengsizlik yozamiz: $-3<7-2x<3$.
2. Uchala qismdan 7 ni ayiramiz: $-10<-2x<-4$.
3. $-2$ ga bo'lamiz, belgilar teskarisiga o'zgaradi: $5>x>2$, ya'ni $2<x<5$.
4. Butun yechimlar: $3$ va $4$. 2 va 5 kirmaydi — uchlari bo'sh doiracha.



**4-MISOL · YOZUVNI ALMASHTIRISH**

1. $|x| \le 0{,}6$ ni qo'sh tengsizlik bilan yozamiz: $-0{,}6 \le x \le 0{,}6$.
2. $-3{,}4 < x < 3{,}4$ ni modul bilan yozamiz: chetlari qarama-qarshi sonlar, demak $|x| < 3{,}4$.
3. $|x| > 5$ ni qo'sh tengsizlik bilan yozib bo'lmaydi: yechim ikkita nur, $x < -5$ yoki $x > 5$.

Hayotda: sog'lom odamning harorati $t$ 36,6 °C dan ko'pi bilan 0,5 °C ga farq qiladi (rasmda).




$$|t - 36{,}6| \le 0{,}5$$
$$36{,}1 \le t \le 37{,}1$$

**5-MISOL · «≥» BELGI, IKKI HOLAT**

$|2x + 1| \ge 5$ tengsizlikni yeching.

1. $2x + 1 \ge 0$ bo'lsin. Sistemani yechamiz:
$$\begin{cases} 2x + 1 \ge 0, \\ 2x + 1 \ge 5 \end{cases} \Rightarrow x \ge 2.$$
2. $2x + 1 < 0$ bo'lsin. Unda $-(2x + 1) \ge 5$, ya'ni $2x + 1 \le -5$. Sistemani yechamiz:
$$\begin{cases} 2x + 1 < 0, \\ 2x + 1 \le -5 \end{cases} \Rightarrow x \le -3.$$
3. Javob: $x \le -3$ yoki $x \ge 2$.




Ikki holatni qisqa yozish ham mumkin: $2x + 1 \ge 5$ yoki $2x + 1 \le -5$.

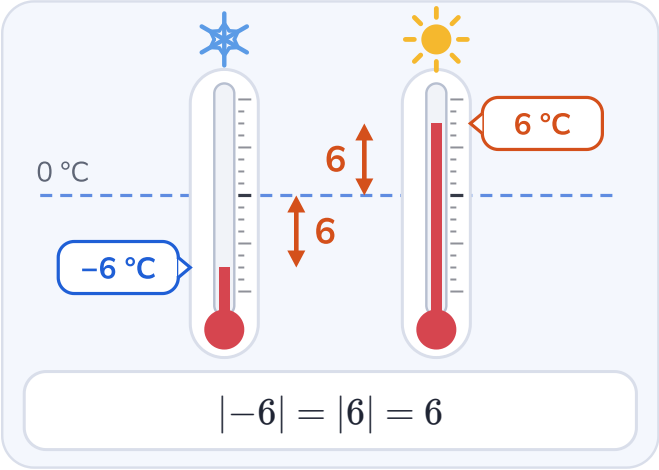
**DIQQAT!**

1. $|x| = 5$: ✗ $x = 5$ ✓ $x = 5$ **yoki** $x = -5$
2. $|x - 3| = -2$: ✗ $x = 1$ yoki $x = 5$ ✓ **ildiz yo'q** — modul manfiy bo'lmaydi.
3. $|x + 1| = 0$: ✗ $x = 1$ yoki $x = -1$ ✓ $x = -1$ — ildiz bitta, chunki $x + 1 = 0$.
4. $|-x| = 3$: ✗ $x = -3$ ✓ $x = 3$ **yoki** $x = -3$ — $|-x| = |x|$.
5. $|x| < 4$: ✗ $x < 4$ ✓ $-4 < x < 4$ — $-4$ va 4 kirmaydi.
6. $|x| > 4$: ✗ $-4 > x > 4$ ✓ $x < -4$ **yoki** $x > 4$
7. $-8 < -2x < -2$ dan: ✗ $4 < x < 1$ ✓ $1 < x < 4$ — $-2$ ga bo'lganda belgilar teskarisiga o'zgaradi.

3-QISM

## Mustahkamlash

1. **(Og'zaki.)** Sonning modulini toping:
   1) $31$; 2) $6{,}2$; 3) $\frac{4}{9}$; 4) $-58$; 5) $-3{,}7$; 6) $-\frac{5}{11}$.




2. Tenglamani yeching:
   1) $|x| = 3{,}5$; 2) $|x| = 0{,}8$; 3) $|x - 2| = 5$;
   4) $|x + 6| = 6$; 5) $|x + 1| = 9$; 6) $|x - 10| = 10$.

3. Tenglamani yeching:
   1) $|x + 9| = 0$; 2) $|x - 5| = 0$; 3) $|3x - 8| = 0$;
   4) $|2 - 5x| = 0$; 5) $|6 + 5x| = 0$; 6) $|4x + 7| = 0$.

4. Tenglamani yeching:
   1) $|2x-7|=3$; 2) $|5x+2|=8$; 3) $\left|\frac{1}{2}x+\frac{1}{4}\right|=\frac{3}{4}$;
   4) $\left|\frac{3}{4}x-1\right|=\frac{1}{2}$; 5) $|6x-11|=5$; 6) $|1{,}5-4x|=3{,}5$.

5. Tenglamani yeching:
   1) $|-x|=4{,}5$; 2) $|-x|=1{,}2$; 3) $|7-x|=7$;
   4) $|2-x|=9$; 5) $|x-8|=3$; 6) $|9-x|=2$.

6. Tengsizlikning yechimlar to'plamini son o'qida tasvirlang:
   1) $|x|<3$; 2) $|x|\le 2$; 3) $|x|\ge 1$; 4) $|x|>4$.

7. Modulli tengsizlikni qo'sh tengsizlik shaklida yozing:
   1) $|x|\le 6$; 2) $|x|<1$; 3) $|x|<4{,}5$; 4) $|x|\le 1{,}8$; 5) $|x|\ge 2$.

8. Qo'sh tengsizlikni bitta modulli tengsizlik shaklida yozing:
   1) $-2{,}6<x<2{,}6$; 2) $-0{,}7\le x\le 0{,}7$; 3) $-5{,}3<x<5{,}3$.

9. Tengsizlikni yeching:
   1) $|5+x|\le 0{,}6$; 2) $|x+7|<0{,}5$; 3) $|4-x|\le\frac{1}{3}$;
   4) $|2-x|<\frac{4}{5}$; 5) $|x-2|\le 3$; 6) $|x-5|\le 1$.

10. Tengsizlikni yeching:
    1) $|2x-9|<7$; 2) $|3x+1|<5$; 3) $|9-2x|\le 5$;
    4) $|7-3x|\le 2$; 5) $|5x-3|<8$; 6) $|8-5x|\le 7$.

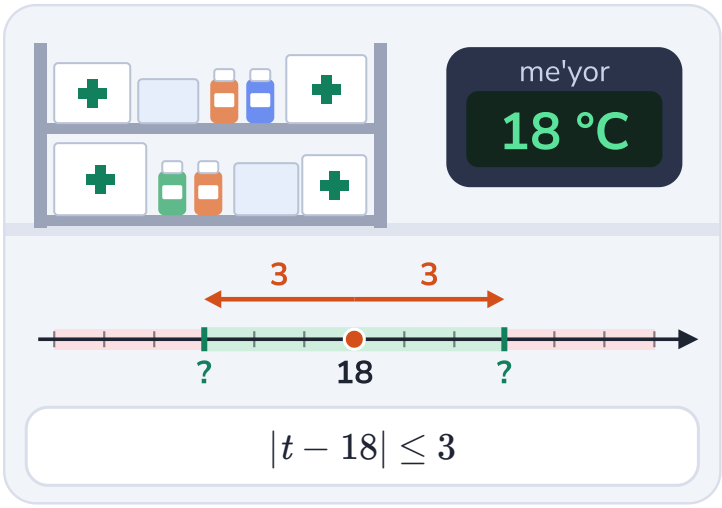
11. Tengsizlikni yeching:
    1) $|x+2|>1{,}5$; 2) $|x-3|\ge 2{,}4$; 3) $|2-x|\ge\frac{1}{4}$;
    4) $|8-x|>\frac{3}{5}$; 5) $|x-6|>2{,}7$; 6) $|6-x|<1{,}5$.

12. Tengsizlikni yeching:
    1) $|3x-7|\ge 4$; 2) $|2x+7|>5$; 3) $|4x-5|>3$;
    4) $|5-2x|\ge 3$; 5) $|4x-9|\le 3$; 6) $|1-4x|\ge 5$.

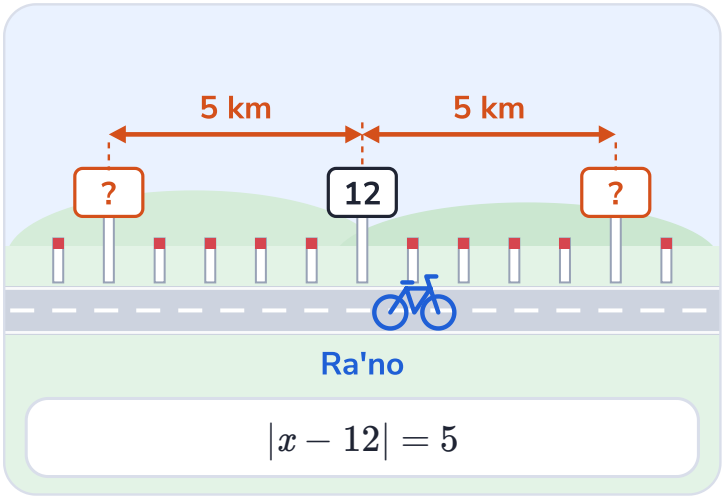
13. $x$ ning tengsizlik bajariladigan barcha butun qiymatlarini toping:
    1) $|3x+5|<7$; 2) $|4x+1|<9$; 3) $|8-3x|\le 2$;
    4) $|5-2x|\le 5$; 5) $|2x-7|\le 3$; 6) $|1-5x|\le 9$.

14. Tengsizlikni yeching:
    1) $|3x - 8| > 4$; 2) $|2x - 11| \le 9$; 3) $|2 - 5x| \le 3$;
    4) $|7 - 4x| \ge 3$; 5) $|2{,}5x - 1| \le 4$; 6) $|6 - 5x| > 4$.

15. Dori-darmon omborida harorat 18 °C atrofida saqlanishi kerak: me'yordan chetlanish 3 °C dan oshmasin, ya'ni $|t - 18| \le 3$.
    a) Ruxsat etilgan harorat oralig'ini toping.
    b) 14 °C, 15 °C, 20,5 °C va 22 °C dan qaysilari ruxsat etilgan?




16. Velosiped yo'li bo'ylab har kilometrda raqamli belgi qo'yilgan. Ra'no 12-km belgisi yonida dam olmoqda.
    a) Ra'nodan roppa-rosa 5 km uzoqlikdagi belgilarni toping ($|x - 12| = 5$).
    b) Ra'nodan 5 km dan uzoq bo'lmagan belgilar qaysi oraliqda ($|x - 12| \le 5$)?




4-QISM

## Mustaqil ish

1. Hisoblang: a) $|-15| + |9|$; b) $|-4{,}2| - |1{,}7|$; c) $|-6| \cdot \left|-\frac{1}{3}\right|$; d) $|3 - 10|$.
2. Tenglamani yeching: a) $|x| = 4{,}2$; b) $|x + 5| = 2$; c) $|4x + 5| = 7$; d) $|7 - 5x| = 0$; e) $|x - 1| = -3$.
3. Tenglamani yeching: a) $|-x| = 1{,}8$; b) $|10 - x| = 3$; c) $\left|\frac{1}{3}x - \frac{1}{2}\right| = \frac{1}{6}$.
4. Yechimlar to'plamini son o'qida tasvirlang: a) $|x| \le 1{,}5$; b) $|x| > 3$.

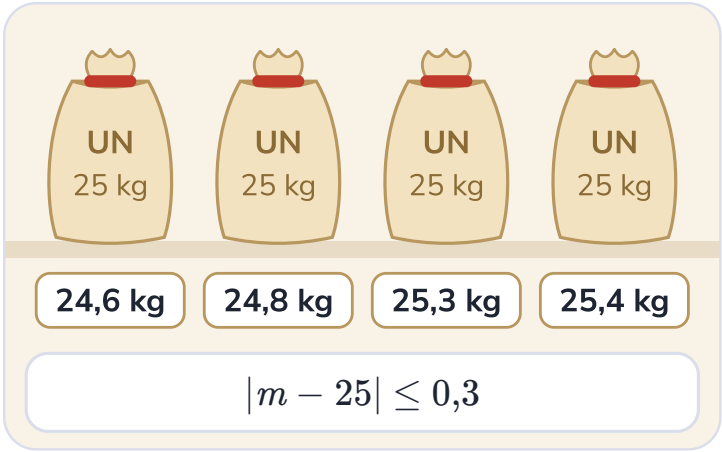
5. a) $|x| < 7$ ni qo'sh tengsizlik shaklida yozing. b) $-0{,}9 \le x \le 0{,}9$ ni modul bilan yozing.

6. Tengsizlikni yeching: a) $|x - 3| \le 4$; b) $|2x + 1| < 9$; c) $|8 - 3x| \le 7$.

7. Tengsizlikni yeching: a) $|x + 2| > 6$; b) $|3x + 4| \ge 5$; c) $|2x + 9| \ge -1$; d) $|x + 8| \le -2$; e) $|4x - 2| \le 0$.

8. Tengsizlikning barcha butun yechimlarini toping: a) $|5x - 4| < 6$; b) $|5 - 2x| \le 3$.

9. Tegirmonda un 25 kg li qoplarga joylanadi. Qop og'irligining me'yordan chetlanishi 0,3 kg dan oshmasligi kerak: $|m - 25| \le 0{,}3$. Tarozida to'rtta qop tortildi: 24,6 kg; 24,8 kg; 25,3 kg; 25,4 kg. Qaysi qoplar talabga javob beradi?




10. Son o'qida 3 nuqtadan 2,5 birlikdan kam masofada yotgan nuqtalarni modulli tengsizlik bilan yozing va uni yeching.

11. **Xatoni toping.** Dilnoza $|3 - x| < 2$ tengsizlikni shunday yechdi: $-2 < 3 - x < 2$; $-5 < -x < -1$; $-5 < x < -1$. Xato qayerda? To'g'ri javob qanday?

12. **Qiyinroq.** $|x - 4| \le 3$ va $|x + 1| > 2$ tengsizliklarning ikkalasini ham qanoatlantiradigan butun sonlarni toping.

5-QISM

# Javoblar

## Mustahkamlash

**1-masala**

1) $31$; 2) $6{,}2$; 3) $\frac{4}{9}$; 4) $58$; 5) $3{,}7$; 6) $\frac{5}{11}$.

**2-masala**

1) $x = 3{,}5$ yoki $x = -3{,}5$. 2) $x = 0{,}8$ yoki $x = -0{,}8$. 3) $x - 2 = 5$ yoki $x - 2 = -5$; $x = 7$ yoki $x = -3$. 4) $x + 6 = 6$ yoki $x + 6 = -6$; $x = 0$ yoki $x = -12$. 5) $x + 1 = 9$ yoki $x + 1 = -9$; $x = 8$ yoki $x = -10$. 6) $x - 10 = 10$ yoki $x - 10 = -10$; $x = 20$ yoki $x = 0$.

**3-masala**

1) $x + 9 = 0$, $x = -9$. 2) $x - 5 = 0$, $x = 5$. 3) $3x - 8 = 0$, $x = 2\frac{2}{3}$. 4) $2 - 5x = 0$, $x = \frac{2}{5}$. 5) $6 + 5x = 0$, $x = -1\frac{1}{5}$. 6) $4x + 7 = 0$, $x = -1\frac{3}{4}$.

**4-masala**

1) $2x - 7 = 3$ yoki $2x - 7 = -3$; $x = 5$ yoki $x = 2$. 2) $5x + 2 = 8$ yoki $5x + 2 = -8$; $x = 1\frac{1}{5}$ yoki $x = -2$. 3) $\frac{1}{2}x + \frac{1}{4} = \frac{3}{4}$ yoki $\frac{1}{2}x + \frac{1}{4} = -\frac{3}{4}$; $x = 1$ yoki $x = -2$. 4) $\frac{3}{4}x - 1 = \frac{1}{2}$ yoki $\frac{3}{4}x - 1 = -\frac{1}{2}$; $x = 2$ yoki $x = \frac{2}{3}$. 5) $6x - 11 = 5$ yoki $6x - 11 = -5$; $x = 2\frac{2}{3}$ yoki $x = 1$. 6) $1{,}5 - 4x = 3{,}5$ yoki $1{,}5 - 4x = -3{,}5$; $x = -0{,}5$ yoki $x = 1{,}25$.

**5-masala**

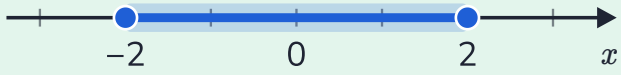
1) $-x = 4{,}5$ yoki $-x = -4{,}5$; $x = -4{,}5$ yoki $x = 4{,}5$. 2) $-x = 1{,}2$ yoki $-x = -1{,}2$; $x = -1{,}2$ yoki $x = 1{,}2$. 3) $7 - x = 7$ yoki $7 - x = -7$; $x = 0$ yoki $x = 14$. 4) $2 - x = 9$ yoki $2 - x = -9$; $x = -7$ yoki $x = 11$. 5) $x - 8 = 3$ yoki $x - 8 = -3$; $x = 11$ yoki $x = 5$. 6) $9 - x = 2$ yoki $9 - x = -2$; $x = 7$ yoki $x = 11$.

**6-masala**

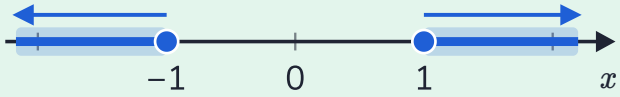
1) $-3 < x < 3$.



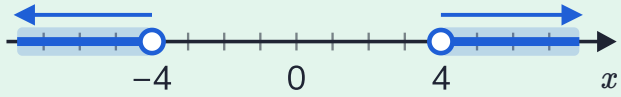

2) $-2 \le x \le 2$.




3) $x \le -1$ yoki $x \ge 1$.




4) $x < -4$ yoki $x > 4$.




**7-masala**

1) $-6 \le x \le 6$. 2) $-1 < x < 1$. 3) $-4{,}5 < x < 4{,}5$. 4) $-1{,}8 \le x \le 1{,}8$. 5) Qo'sh tengsizlik bilan yozib bo'lmaydi: yechim ikkita nur, $x \le -2$ yoki $x \ge 2$.

**8-masala**

1) $|x| < 2{,}6$; 2) $|x| \le 0{,}7$; 3) $|x| < 5{,}3$.

**9-masala**

1) $-0{,}6 \le 5 + x \le 0{,}6$; $-5{,}6 \le x \le -4{,}4$. 2) $-0{,}5 < x + 7 < 0{,}5$; $-7{,}5 < x < -6{,}5$. 3) $-\frac{1}{3} \le 4 - x \le \frac{1}{3}$; $-4\frac{1}{3} \le -x \le -3\frac{2}{3}$; $-1$ ga bo'lamiz, belgilar teskarisiga o'zgaradi: $3\frac{2}{3} \le x \le 4\frac{1}{3}$. 4) $-\frac{4}{5} < 2 - x < \frac{4}{5}$; $-2\frac{4}{5} < -x < -1\frac{1}{5}$; $-1$ ga bo'lamiz, belgilar teskarisiga o'zgaradi: $1\frac{1}{5} < x < 2\frac{4}{5}$. 5) $-3 \le x - 2 \le 3$; $-1 \le x \le 5$. 6) $-1 \le x - 5 \le 1$; $4 \le x \le 6$.

**10-masala**

1) $-7 < 2x - 9 < 7$; $2 < 2x < 16$; $1 < x < 8$. 2) $-5 < 3x + 1 < 5$; $-6 < 3x < 4$; $-2 < x < 1\frac{1}{3}$. 3) $-5 \le 9 - 2x \le 5$; $-14 \le -2x \le -4$; $-2$ ga bo'lamiz, belgilar teskarisiga o'zgaradi: $2 \le x \le 7$. 4) $-2 \le 7 - 3x \le 2$; $-9 \le -3x \le -5$; $-3$ ga bo'lamiz, belgilar teskarisiga o'zgaradi: $1\frac{2}{3} \le x \le 3$. 5) $-8 < 5x - 3 < 8$; $-5 < 5x < 11$; $-1 < x < 2\frac{1}{5}$. 6) $-7 \le 8 - 5x \le 7$; $-15 \le -5x \le -1$; $-5$ ga bo'lamiz, belgilar teskarisiga o'zgaradi: $\frac{1}{5} \le x \le 3$.

**11-masala**

1) $x+2>1{,}5$ yoki $x+2<-1{,}5$; $x<-3{,}5$ yoki $x>-0{,}5$. 2) $x-3\geq 2{,}4$ yoki $x-3\leq -2{,}4$; $x\leq 0{,}6$ yoki $x\geq 5{,}4$. 3) $2-x\geq \frac{1}{4}$ yoki $2-x\leq -\frac{1}{4}$; $x\leq 1\frac{3}{4}$ yoki $x\geq 2\frac{1}{4}$. 4) $8-x>\frac{3}{5}$ yoki $8-x<-\frac{3}{5}$; $x<7\frac{2}{5}$ yoki $x>8\frac{3}{5}$. 5) $x-6>2{,}7$ yoki $x-6<-2{,}7$; $x<3{,}3$ yoki $x>8{,}7$. 6) $-1{,}5<6-x<1{,}5$; $-7{,}5<-x<-4{,}5$; $-1$ ga bo'lamiz, belgilar teskarisiga o'zgaradi: $4{,}5<x<7{,}5$.

**12-masala**

1) $3x-7\geq 4$ yoki $3x-7\leq -4$; $x\leq 1$ yoki $x\geq 3\frac{2}{3}$. 2) $2x+7>5$ yoki $2x+7<-5$; $x<-6$ yoki $x>-1$. 3) $4x-5>3$ yoki $4x-5<-3$; $x<\frac{1}{2}$ yoki $x>2$. 4) $5-2x\geq 3$ yoki $5-2x\leq -3$; $x\leq 1$ yoki $x\geq 4$. 5) $-3\leq 4x-9\leq 3$; $6\leq 4x\leq 12$; $1\frac{1}{2}\leq x\leq 3$. 6) $1-4x\geq 5$ yoki $1-4x\leq -5$; $x\leq -1$ yoki $x\geq 1\frac{1}{2}$.

**13-masala**

1) $-7<3x+5<7$; $-12<3x<2$; $-4<x<\frac{2}{3}$. Butun yechimlar: $-3;\ -2;\ -1;\ 0$. 2) $-9<4x+1<9$; $-10<4x<8$; $-2\frac{1}{2}<x<2$. Butun yechimlar: $-2;\ -1;\ 0;\ 1$. 3) $-2\leq 8-3x\leq 2$; $-10\leq -3x\leq -6$; $-3$ ga bo'lamiz, belgilar teskarisiga o'zgaradi: $2\leq x\leq 3\frac{1}{3}$. Butun yechimlar: $2;\ 3$. 4) $-5\leq 5-2x\leq 5$; $-10\leq -2x\leq 0$; $-2$ ga bo'lamiz, belgilar teskarisiga o'zgaradi: $0\leq x\leq 5$. Butun yechimlar: $0;\ 1;\ 2;\ 3;\ 4;\ 5$. 5) $-3\leq 2x-7\leq 3$; $4\leq 2x\leq 10$; $2\leq x\leq 5$. Butun yechimlar: $2;\ 3;\ 4;\ 5$. 6) $-9\leq 1-5x\leq 9$; $-10\leq -5x\leq 8$; $-5$ ga bo'lamiz, belgilar teskarisiga o'zgaradi: $-1\frac{3}{5}\leq x\leq 2$. Butun yechimlar: $-1;\ 0;\ 1;\ 2$.

**14-masala**

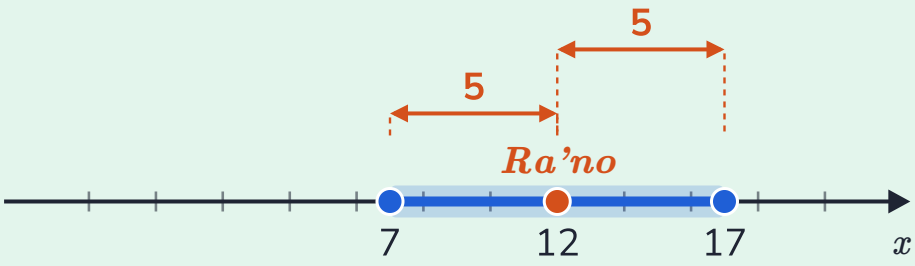
1) $3x - 8 > 4$ yoki $3x - 8 < -4$; $x < 1\frac{1}{3}$ yoki $x > 4$. 2) $-9 \le 2x - 11 \le 9$; $2 \le 2x \le 20$; $1 \le x \le 10$. 3) $-3 \le 2 - 5x \le 3$; $-5 \le -5x \le 1$; $-5$ ga bo'lamiz, belgilar teskarisiga o'zgaradi: $-\frac{1}{5} \le x \le 1$. 4) $7 - 4x \ge 3$ yoki $7 - 4x \le -3$; $x \le 1$ yoki $x \ge 2\frac{1}{2}$. 5) $-4 \le 2{,}5x - 1 \le 4$; $-3 \le 2{,}5x \le 5$; $-1{,}2 \le x \le 2$. 6) $6 - 5x > 4$ yoki $6 - 5x < -4$; $x < \frac{2}{5}$ yoki $x > 2$.

**15-masala**

a) $-3 \le t - 18 \le 3$; $15 \le t \le 21$, ya'ni 15 °C dan 21 °C gacha. b) 15 °C va 20,5 °C (15 — chegara, u ham kiradi); 14 °C va 22 °C ruxsat etilmagan.

**16-masala**

a) $x - 12 = 5$ yoki $x - 12 = -5$: 17-km va 7-km belgilari. b) $-5 \le x - 12 \le 5$; $7 \le x \le 17$: 7-km dan 17-km gacha.




## Mustaqil ish

**1-masala**

a) $15 + 9 = 24$; b) $4{,}2 - 1{,}7 = 2{,}5$; c) $6 \cdot \frac{1}{3} = 2$; d) $|-7| = 7$.

**2-masala**

a) $x = 4{,}2$ yoki $x = -4{,}2$. b) $x + 5 = 2$ yoki $x + 5 = -2$; $x = -3$ yoki $x = -7$. c) $4x + 5 = 7$ yoki $4x + 5 = -7$; $x = \frac{1}{2}$ yoki $x = -3$. d) $7 - 5x = 0$, $x = 1\frac{2}{5}$. e) Ildiz yo'q: modul manfiy son bo'lmaydi.

**3-masala**

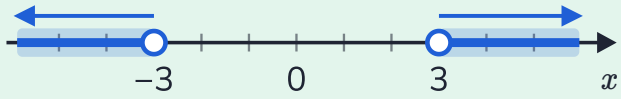
a) $-x = 1{,}8$ yoki $-x = -1{,}8$; $x = -1{,}8$ yoki $x = 1{,}8$. b) $10 - x = 3$ yoki $10 - x = -3$; $x = 7$ yoki $x = 13$. c) $\frac{1}{3}x - \frac{1}{2} = \frac{1}{6}$ yoki $\frac{1}{3}x - \frac{1}{2} = -\frac{1}{6}$; $x = 2$ yoki $x = 1$.

**4-masala**

a) $-1{,}5 \le x \le 1{,}5$.




b) $x < -3$ yoki $x > 3$.




**5-masala**

a) $-7 < x < 7$; b) $|x| \le 0{,}9$.

**6-masala**

a) $-4 \le x - 3 \le 4$; $-1 \le x \le 7$. b) $-9 < 2x + 1 < 9$; $-10 < 2x < 8$; $-5 < x < 4$. c) $-7 \le 8 - 3x \le 7$; $-15 \le -3x \le -1$; $-3$ ga bo'lamiz, belgilar teskarisiga o'zgaradi: $\frac{1}{3} \le x \le 5$.

**7-masala**

a) $x + 2 > 6$ yoki $x + 2 < -6$; $x < -8$ yoki $x > 4$. b) $3x + 4 \ge 5$ yoki $3x + 4 \le -5$; $x \le -3$ yoki $x \ge \frac{1}{3}$. c) Istalgan son: modul hech qachon manfiy emas, u doim $-1$ dan katta. d) Yechim yo'q: modul manfiy son bo'lmaydi. e) $4x - 2 = 0$, $x = \frac{1}{2}$.

**8-masala**

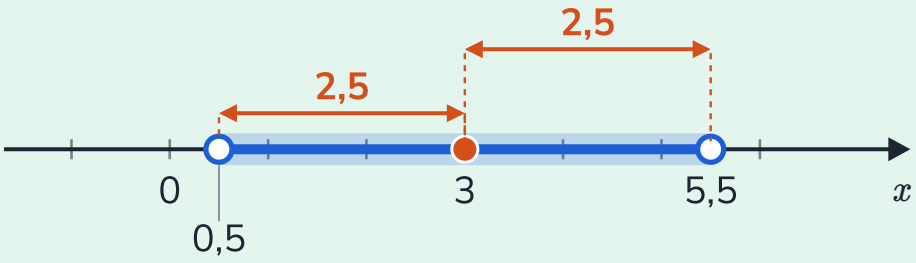
a) $-6 < 5x - 4 < 6$; $-2 < 5x < 10$; $-\frac{2}{5} < x < 2$. Butun yechimlar: $0;\ 1$. b) $-3 \le 5 - 2x \le 3$; $-8 \le -2x \le -2$; $-2$ ga bo'lamiz, belgilar teskarisiga o'zgaradi: $1 \le x \le 4$. Butun yechimlar: $1;\ 2;\ 3;\ 4$.

**9-masala**

$-0{,}3 \le m - 25 \le 0{,}3;\ 24{,}7 \le m \le 25{,}3$. Talabga javob beradi: 24,8 kg va 25,3 kg (25,3 — chegara, kiradi).

**10-masala**

$|x - 3| < 2{,}5;\ -2{,}5 < x - 3 < 2{,}5;\ 0{,}5 < x < 5{,}5$.




**11-masala**

Uchinchi qadamda Dilnoza faqat $x$ oldidagi minusni olib tashlagan. Aslida uchala qismni $-1$ ga bo'lish kerak: chetdagi sonlar 5 va 1 bo'ladi, belgilar teskarisiga o'zgaradi: $5 > x > 1$, ya'ni $1 < x < 5$. Tekshirish: $x = -3$ da $|3 - (-3)| = 6$, 6 esa 2 dan kichik emas — demak Dilnozaning javobi noto'g'ri.

**12-masala**

$|x - 4| \le 3$: $1 \le x \le 7$. $|x + 1| > 2$: $x < -3$ yoki $x > 1$. Ikkalasi ham bajariladi: $1 < x \le 7$. Butun sonlar: $2;\ 3;\ 4;\ 5;\ 6;\ 7$.

